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Conceptual Design of a UAV-UGV Autonomous Collaborative Robot System

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***Abstract*—Collaborative robots (cobots) are in the focus of attention since many years being applied widely in different fields of the industry and national economies. In classical meaning and content of the word, cobots are cooperating with human operators in shared space working for the common purpose of the effective production of goods or other products. This article will extend the meaning of the word cobot to the collaboration of two autonomous robot systems, such as ground and air robots working together in close and shared space to execute different safety and security missions.**

Keywords—collaborative, robots, autonomous systems, UAV/UAV, UGV

I. INTRODUCTION

Recent collaborative robots (cobots) and robot systems are designed to work in the friendly and well-known environment, working together with human operators in joint missions.

The cobots are often substituting some missing human resources in the manufacturing and national economies. Moreover, use of cobots makes the production less expensive, more reliable and predictable.

In many cobot applications the level of the complexity of the tasks delivered to cobots is still not so high; however, there are several privileges of the cobot applications like high level of accessibility, high power in actions, safe functioning, fast movements allowing high production rate in the production process.

Collaborating robot systems are mostly pre-programmed ones with given level of autonomy. In shared space with humans they are able to execute their missions they designed and prepared for. One of the major weakness of the industrial cobots is the lack of autonomy in its very broad meaning.

The cobots can be used both for the private, non-state and for the state purposes. The private, non-state use of cobots is widely spread in the manufacturing and in national economies.

The state applications of cobots can cover both military and non-military applications. The non-military applications of cobots are in disaster relief and in industrial catastrophe missions, when the cobot mission replaces human activity in the dirty, dull and dangerous missions when humans are under threat of different dangerous events.

The military applications of cobots today are at quiet low level executing mostly logistical tasks. However, many challenges at the battlefield can be handled by autonomous cobots.

II. MOTIVATION AND PROBLEM FORMULATION

The main goal of this paper is to formulate the conceptual design of the autonomous cobot system based upon two subsystems like ground autonomous unmanned vehicle (UGV) and the autonomous unmanned aerial vehicle system (UAV) collaborating in joint ground and air missions.

The basic idea behind these co-operating system is a synergistic and holistic approach in cobot system design eliminating weaknesses and disadvantages of both of the collaborating systems and strengthening advantages of both of the co-operating systems.

In this article, in the framework of the international cobot system design competition, the main purpose is to solve different problems of the 2D and 3D path design and navigation, re-charge of batteries and battery packs; searching, finding and gripping of ground objects based on smart vision systems.

The scenario of the cobot system design is described to be solved in a cobot competition arena (CAA) designed for the unique purpose of the competition.

III. LITERATURE REVIEW

Undoubtedly, the collaborative robots (cobots) had have gained unavoidable role in different fields of the industrial manufacturing.

In [1] a thorough literature review is shown to give an overview of the recent state-of-the-art application of the cobots, focusing on Industry 4.0 challenges of the past decade when a human and a robot work together in the joint work cells. The use of cobots was always a challenge due to their human acceptance, or negligence.

The ‘human-robot’ collaboration acceptance model was elaborated and introduced in [2] leaning on data gained from different countries. This study has set up a new cobot acceptance model able to express also the cross-cultural differences.

The more complex industrial scenario is described in [3], when one has to investigate interactions between cobots, or between human and robots.

This article gives new content to the human-robot interactions enhancing robot performances via better understanding of the human. Signals like facial animation, hand gesture and arm motion.

The modern mass-production often require to run collaborating assembly lines operating simultaneously to multiple production rate. In the production different kinds of asymmetries in the human-robot interactions might appear.

In [4] for production line balancing an optimization method has been introduced. In the study of the European Parliament [5] risks and opportunities have been evaluated for cobot market development.

In spite of the deep penetration of the cobots into national economies, cobots can be used very effectively in shared spaces.

Moreover, in some disaster management tasks to strengthen efficacy of the first aid delivered to people snake robots were used [6].

The article [7] focuses on industrial cobots evaluating challenges of the manufacturers when to implement them. Several factors have been involved when industrial applicability of the cobots was evaluated.

According to [8], also in the modern production and manufacturing systems different human factors like ergonomics, mental workload, trust, acceptance and usability are crucially important to take into consideration when to design human operators' workplace.

Recently the automotive industry is a segment of the national economies, which is widely lean on cobots in solution of welding tasks reducing or eliminating human interactions in the production [9].

The control and monitoring of cobots is a crucial issue to ensure safety and security both of the workcell and the human operator. For this purpose, in [10] a new software system is elaborated to control a cobot's gripper position.

The competitiveness of any cobot solution is a core criterion for all applications. In [11] a case study is shown for an automotive supplier priorities based upon skills distribution between cobots and operators.

The article [12] addresses some useful results of the interview made to evaluate human satisfaction with cobots in manufacturing and production.

Cobots are also used very effectively in construction industry. However, it is important to be familiar with the trends in construction cobot developments and to have precise and reliably forecast how the market of construction cobots will develop. In this field, article [13] gives trends in safe construction cobot development.

Any substitution of human operator using cobots in production process is motivated by different reasons. In article [14] an example is given to calculate efficiency and cost effectiveness of the cobot based production system leading to an inevitable decision to substitute human.

In the case of cooperative robot systems, the movement accuracy of individual robots must be maintained even in the case of changing mass. [15] A change in load cannot cause significant deviations in the system, because this hinders cooperation and can sometimes be a source of accidents.

Significant research reports and case studies [16] [17] have also been published on cooperative robot collaboration.

IV. The UAV-UGV Collaborative Robot System

In this regard, the autonomous UAV-UGV system serves the purpose to conduct an aerial recconnaissance using UAV to find different objects made of different materials and colours on the arena ground, to identify them, to pick them up with grippers, and to carry them into designated area. The more objects are found and placed ino designated area the best results are gained.

V. The autonomous UAV (AUAV) System

The autonomous multirotor AUAV is of any kind equipped with electric propulsion system. The UAV MTOW is 2 kg, including its payload.

The AUAVs are also prepared for the wireless recharge of the batteries they use aboard. Before the first run the AUAV is put into its landing zone (LZ) segregated inside the arena manually.

After all flight missions, or after any mission aborted the UAV shall return to its LZ and shut down its propulsion. If the AUAV is leaving or reaching the segregated LZ it shall be signalized with flashing navigation lights during 2.5 seconds with 2Hz of frequency.

Further, any signalization provided by the AUAV means flashing navigation lights during 2.5 seconds with 2Hz of frequency.

The AUAV shall own proper gripping systems able to pick up targets from the floor. In case of necessity, the AUAV is prepared for change of the gripping system it owns. Any gripper change manoeuvre shall be conducted in the LZ area.

VI. The autonomous UGV (AUGV) System

The autonomous AGV is a multipurpose ground rover with omnidirectional drive system carrying battery pack for wireless recharge of the batteries of the AUAVs.

It also represents a catapult system able to throw target devices from its magazine installed aboard into the arena in three different directions previously selected by the teams for the distance of 2 metres, as the minimum.

The magazine bay is normally closed with doors, and opens at proper signalization prior to throw targets onto the arena floor.

In the AUGV default home position, in the rover corner (RC), it rests in the dock supplied with 230V/50Hz electricity serving purpose of the wireless recharging of the battery pack of the AGV.

The maximum weight of the rover is 10 kg with maximums of 40*40*40 centimetres of length, with and height, respectively.

The top of the rover is a temporary landing zone of the AUAVs serving simultaneously the wireless recharge of the AUAV batteries used aboard.

The LZ mounted on the top of the AUGV might have sizes differing from the rovers' main sizes, but shall not exceed of the size of 50*50 cm.

Signalization provided by the AUGV means both the siren and flashing lights active during 2.5 seconds with 2Hz of frequency.

VII. The Competition Arena (CA)

The indoor Competition Arena (CA) has natural and artificial lighting, depending of the time.

The CA has sizing of 7*7 metres. The floor inside of the CA is covered with commercial OSB plates.

The CA has 30 cm high walls made of OSB plates, too, continued in 1 metres high plastic nets.

The top of the arena is also covered with light fishing plastic nets to avoid any unwanted and uncontrolled runaway flight of the UAV.

At 60 cm there is metallic strip of 2 cm width glued onto the OSB plate. LZ: a squared area with 60*60 cm of dimensions.

At 60 cm there is red plastic strip of 2 cm width glued onto the standard OSB plate. Warehouses (W1, W2 and W3): a squared area with 70*70 cm of dimensions.

At 70 cm there are black plastic stripes of 2 cm width glued onto the OSB plate.

The Rendez-Vous Area (RA): is a squared area of the CA sized with 1.5*1.5 metres, and designated on the floor with blue coloured plastic strip of 2 cm width glued onto the OSB plate.

Point ‘A’ is a square-shaped centre of the RA sized with 0.7*0.7 metres, and encircled with the Al-strip of 2 cm width.

Touch points (TP): B and C are touch points on the wall of the CA located at 15 cm height as a square with 10*10 cm sizes. Any reach of any TP shall be signalized like AUGV used to do it.

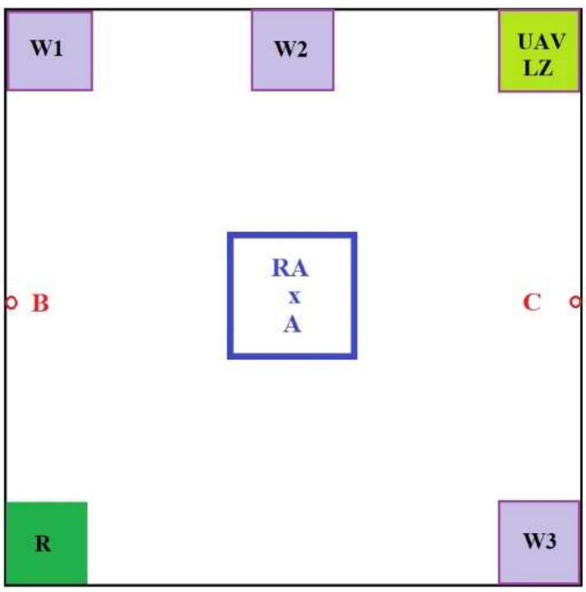

Fig. 1. The Competition Arena from top view

VIII. The Scenario and Competition Tasks to Be Solved

The cobot system designed for the competition has to solve following tasks:

A. *Task No1*

The AUAV takes off for its first flight mission and starts to hover over the LZ for minimum 10 seconds at 0.5 m of altitude.

After, the AUAV leaves the LZ to the RA. The leave of the LZ shall be signalized. The UAV reaches the RA, which shall be signalized.

After, the AUAV returns to its LZ, lands and shuts down its propulsion. The reach of the LZ is signalized by the UAV.

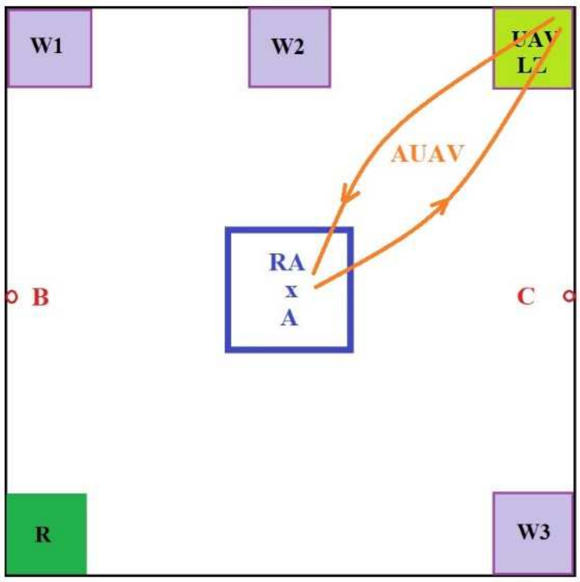

Fig. 2. Representing the first task (Task No1) in CA

B. *Task No2*

The AGV conducts his first autonomous run to reach firstly the A point of the RA with proper signalization.

At point A the AGV continues to manoeuvre to the next TP, with touching the wall at point B with signalization.

After leaving point B the UGV starts to manoeuvre to point C via point A.

On the way from point B to point C, at point A, the AGV stops, and randomly fires out the target items carried by it. The reach of point A is signalized.

Further, the AGV leaves point A and continues his run to point C. The AGV touches point C, which is signalized.

After this, the AUGV returns back to the RC area. The reach of the RC is signalized by the AGV.

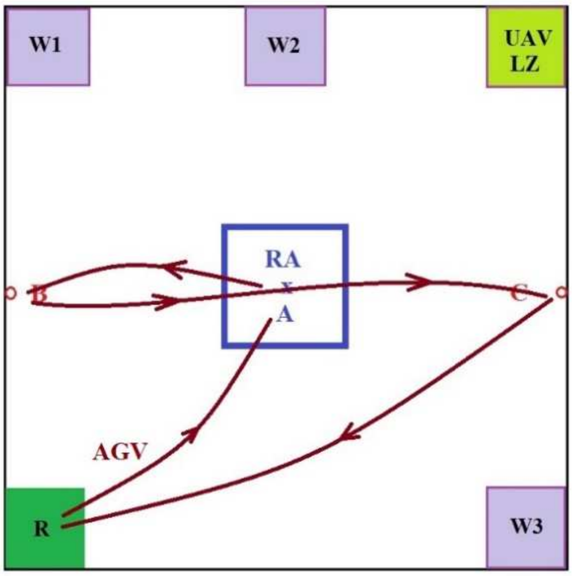

Fig. 3. Representing the second task (Task No2) in CA

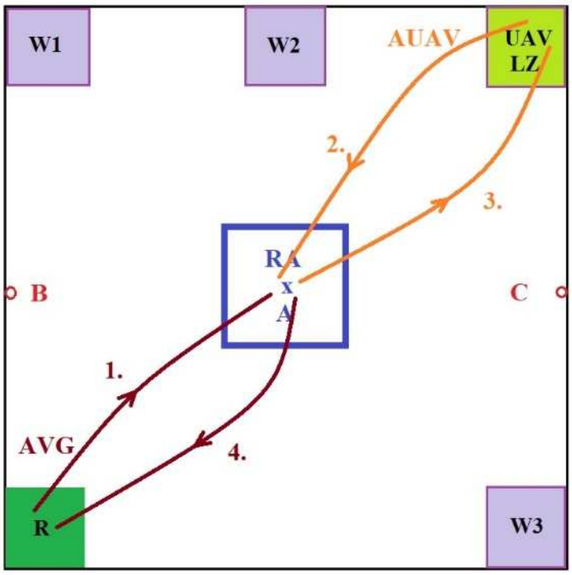

Fig. 4. Representing the fourth task (Task No4) in CA

C. *Task No3*

The AUAV takes off, and conduct search flight missions to find and identify targets thrown onto the floor.

The found targets shall be delivered to the warehouses following the rule: target 'a' to W1, target 'b' to W2 and target 'c' to 'W3'.

D. *Task No4*

At certain, random execution time derived by the jury a wireless fast recharge process of the AUAV batteries shall be activated, which takes place for 5 minutes.

For the signal 'Recharge', firstly, any mission executed by the AUAV shall be interrupted, and a new 'Hover' flight regime shall be set at altitude 1 m for times needed.

The moment when the AGV starts to leave the charger-dock with signalization the AUAV starts to find the AGV in the RA area and its moving LZ to land on it. The AGV streams to point 'A'.

When entering the RA field signalization is activated. At 4:45 seconds of the wireless recharge process the two vehicles disengage.

Firstly, the AUAV takes off for 1 m altitude and returns back to the mission it previously executed.

Secondly, after the AUAV takes off, the AGV returns to RC area with signalization when to dock onto the charger.

E. *Task No5*

In case of necessity if any AUAV runs out of electrical energy on the request of the competing teams further recharge processes might be activated.

IX. Discussions and Closing Remarks

In this paper, the basics of cooperative robot competition are defined. The details and fine-tuning of the competition will take place after the conference. The competition will be announced based on the finalized rules. The scoring system is currently under development.

We trust that the cooperative robot competition will serve the development of this discipline.

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References

[1] M. Knudsen and J. Kaivo-Ojaa, "Collaborative Robots: Frontiers of Current Literature" Journal of Intelligent Systems: Theory and Applications, Vol. 3(2), pp. 13–2, 2020.

[2] C. Bröhl, J. Nelles, C. Brandl, A. Mertens and V. Nitsch "Human-Robot Collaboration Acceptance Model: Development and Comparision for Germany, Japan, China and USA", International Journal of Social Robotics, Vol11, pp. 709-726, 2019.

[3] N. Kakade "Collaborative Robots and Human Robot Interaction: A Review on Two Revolutions of Industry 4.0", Vol11(01), 2023.

[4] A. Keshvarparast, O. Battaia, A. Pirayesh and d. Battini "Considering physical workload and workforce diversity in a Collaborative Asembly Line Balamncing (C-ALB) optimization model", IFAC PapersOnline, 55-10, pp. 157-192, 2022.

[5] E. Gambao, Analysis exploring risks and opportunities linked to the use of collaborative industrial robots in Europe, European Parliamnet Research Center, ISBN: 978-92-848-0799-4, p. 42, 2023.

[6] S.K.R. Moosavi, M.H. Zafar and F. Sanfilippo "Collaborative robots (cobots) for disaster risk resilience: a framework for swarm of snake robots in delivering first aid in emergency situations", Frontiers in Robotics and AI, Vol11, pp. 1-12, 2024.

[7] S.K.L. Anderson, A. Granlund, J. Bruch and M. Hedelind „Experienced Challenges When Implementing Collaborative Robot Applications in Asembly Operations", Int Journal of Automation Technology, Vol15, No5, pp. 678-688, 2021.

[8] M. Faccio et al. „Human factors in cobot era: a review of odern production systems features", Journal of Intelligent Manufacturing, Vol34, pp. 85-106, 2023.

[9] M. Polonara, A. Romagnoli, G. Biancini and L. Carbonari „Introduction of Collaborative Robotics in the Production of Automotive Parts: A Case Study", Machines, Vol12, pp. 11-12, 2024.

[10] W. Łabuński and A. Burghardt, „Software for the Control and Monitoring of Work of a Collaborative Robot", Journal of Automation, Mobile Robotics and Intelligent Systems, Vol15, No3, pp. 29-36, 2021.

[11] M. Vido, G. Scur, A.A: Massote and F. Lima „The impact of the collaborative robot on competitive priorities: case study of an automotive supplier", *Gestão & Produção*, Vol27 (4), e5358, pp. 1-21, 2020

[12] P.Gustavsson and A. Syberfeldt „The Industry's Perspective of Suitable Tasks for Human-Robot Collaboration in Assembly Manufacturing", IOP Conf. Series: Materials Science and Engineering, 1063, p. 1-7, 2020.

[13] B. Alan, C. Glenda and G. Mattthias "Towards Human-Robot Collaboration in Construction: Current Cobot Trends and Forecasts", Construction Robotics, 6, pp. 209-220, 2022.

[14] Y. Cohen and S. Shoval „A New Cobot Deployment Strategy in Manual Assembly Stations: Countering the Impact of Absenteeism", IFAC-PaapersOnline, Vol53, Issue 2, pp. 10275-10278, 2020.

[15] Simge Unsal, Murat Cenk Yilmaz, Yasin Yagin, Hatice Kubra Oner, Volkan Sezer: A New Velocity Planning Method for Autonomous Robots with Varying Mass. ACTA Polytechnica Hungarica Vol19. ISSUE No9. pp.157-177. 2022. ISSN1785-8860

[16] Leonid Sheremetov, Alexander Smirnov, Nikolay Teslya: Dynamic Multi-Robot Coalition Formation: Precision Agriculture Case Study. ACTA Polytechnica Hungarica Vol19. ISSUE No10. pp.221 – 242. 2022. ISSN1785-8860

[17] Mustafa Özdemir and Sitki Kemal Ider: Integrated Force/Motion Trajectory Design of Parallel Robots for Singularity Robustness during Contact Tasks. ACTA Polytechnica Hungarica Vol20. pp41-61. ISSUE No2. 2023. ISSN1785-8860.